



Hamilton Hand-off Moisture Reduction Unit

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August 21, 2026

AMGEN

About Me :)



- Mechanical Engineering (Medical Product Design) @ MIT
- Design Minor
- 2026, just graduated!
- From Los Angeles
- Things I do for fun: Rock climbing, nail art, travel, photography
- Clubs I was in: MakeMIT, MIT Gala, AXO/Panhel Exec



My Projects This Summer!

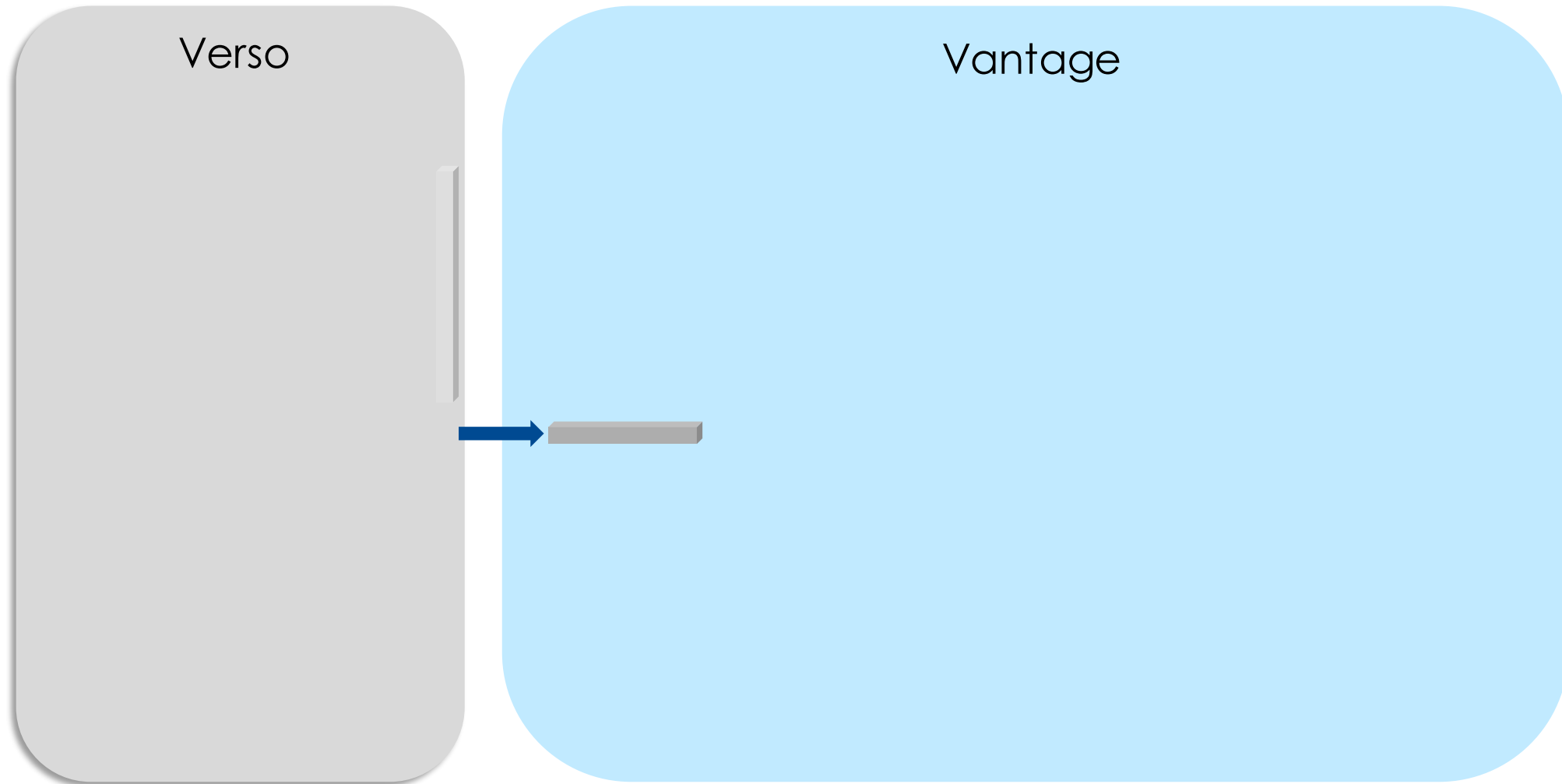
1. Hamilton Hand-off Load Bearing
2. Hamilton Hand-off Moisture Reduction
3. Extension Modules
4. Echo Feet Holders
5. Festo Mounting Plates

} Side quests!

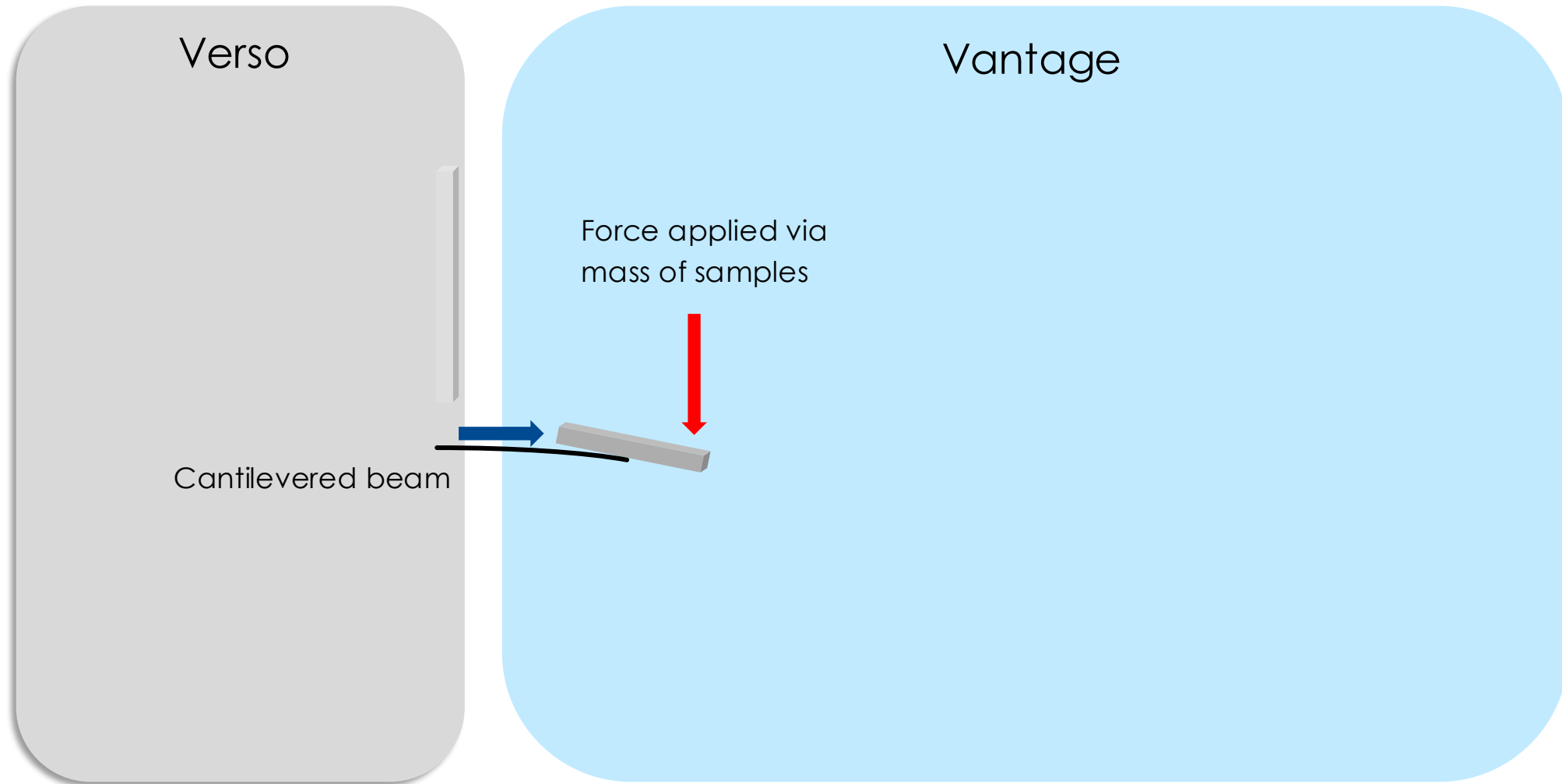
Hamilton Hand-off



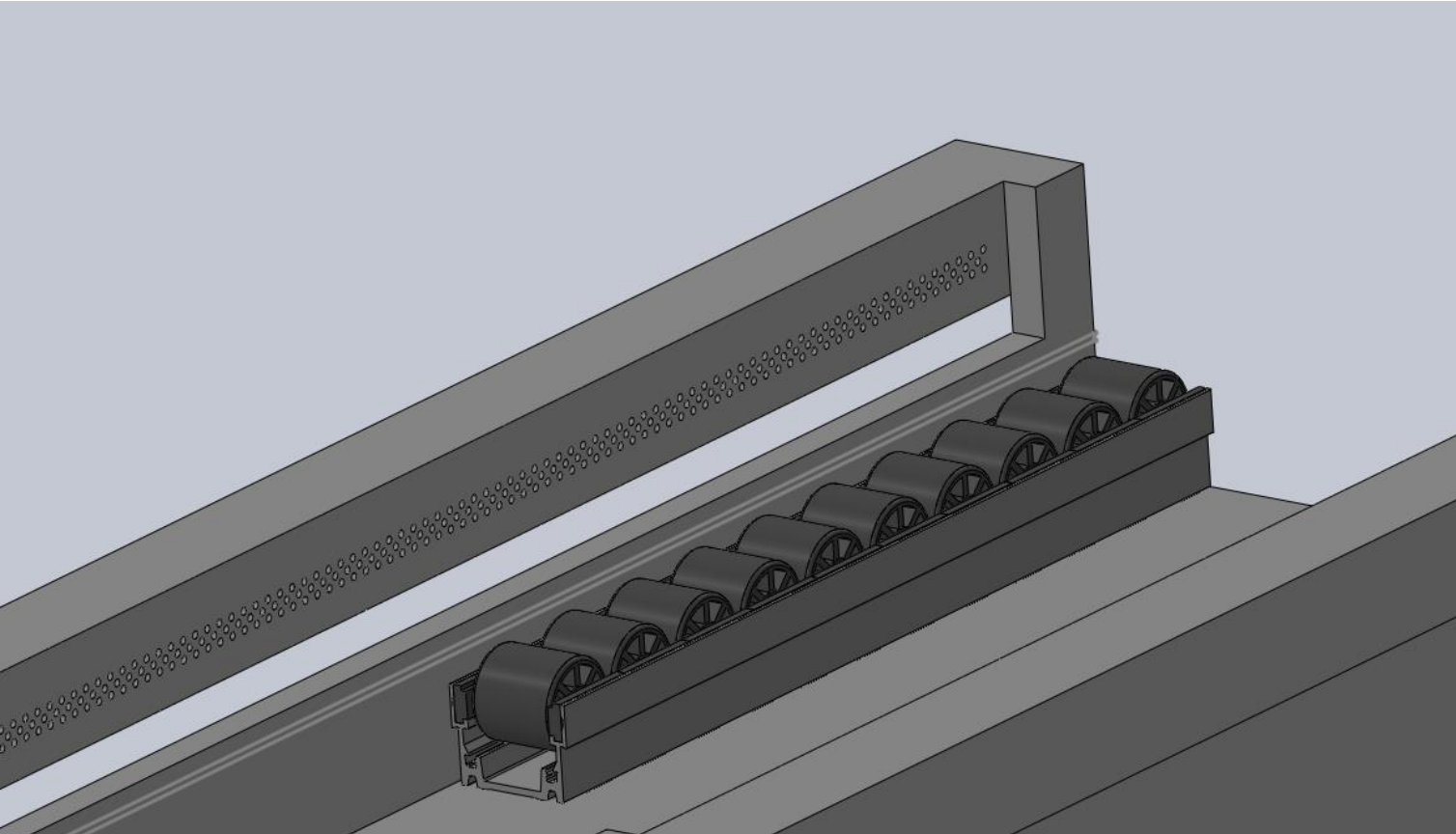
1. Hamilton Hand-off Load Bearing



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1. Solution – Bearings

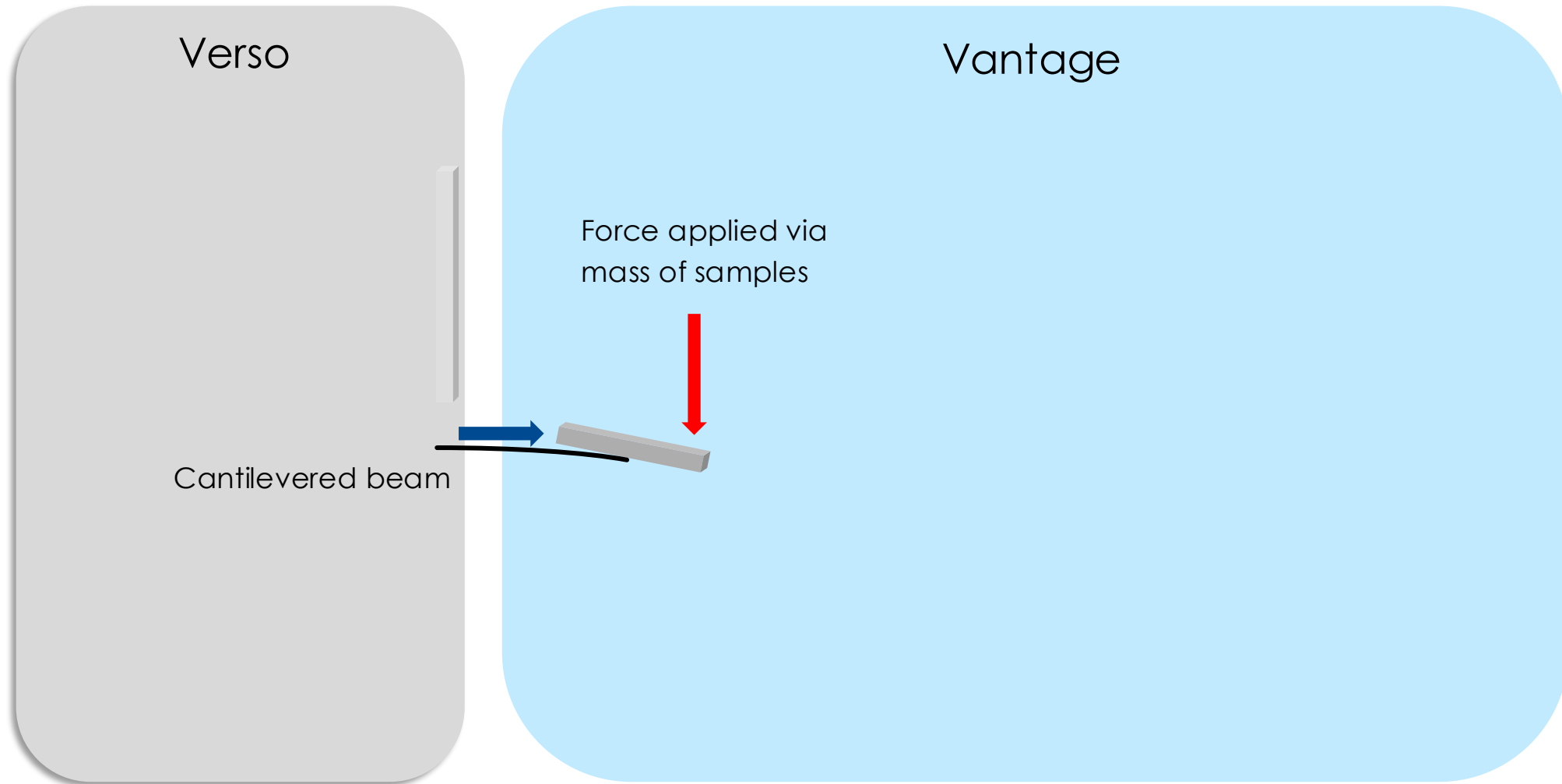


Create a platform for the arm to move on.

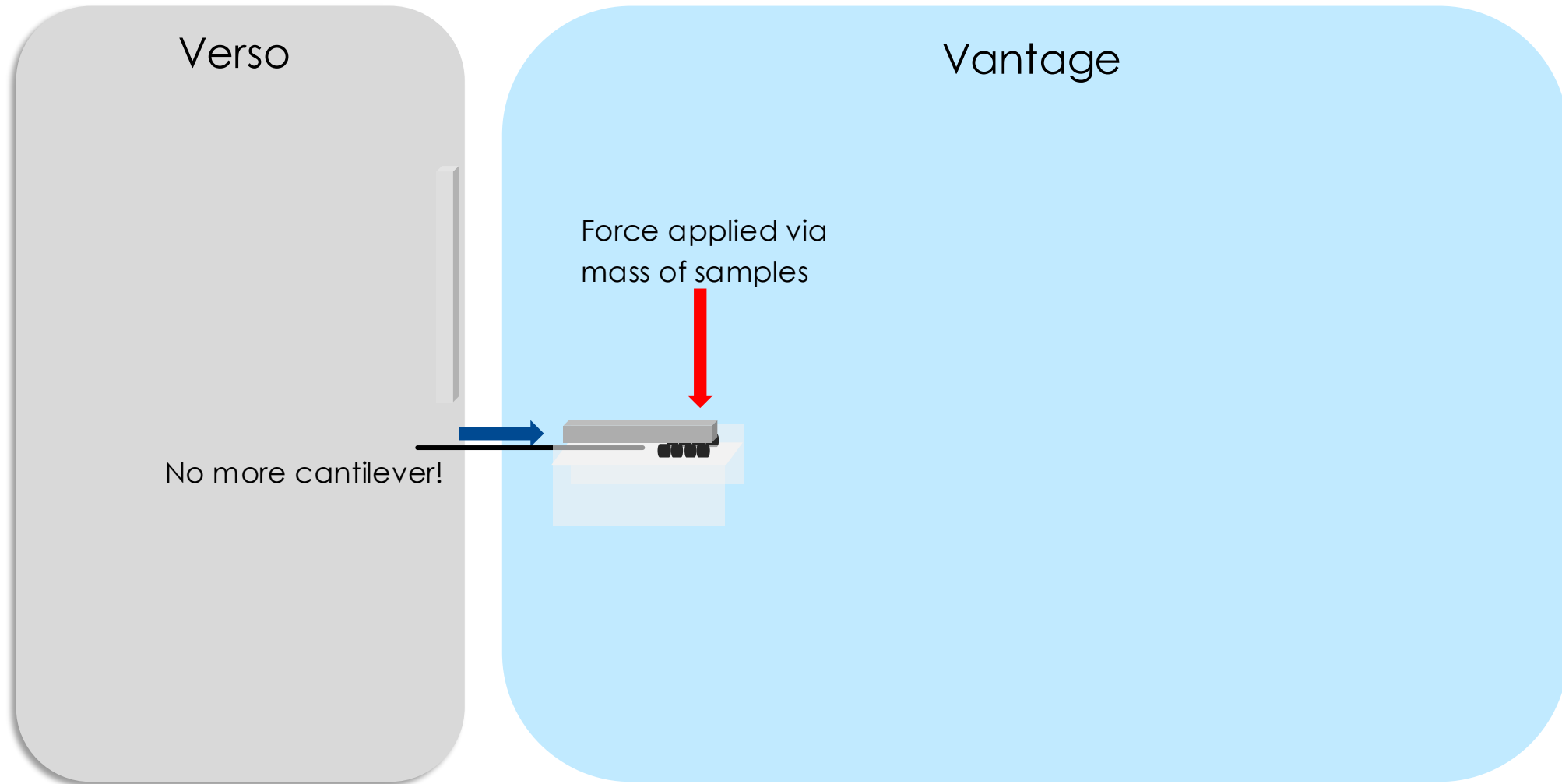
To reduce friction and ensure smooth movement, ball bearings on the surface of the device

Switched to premanufactured rollers

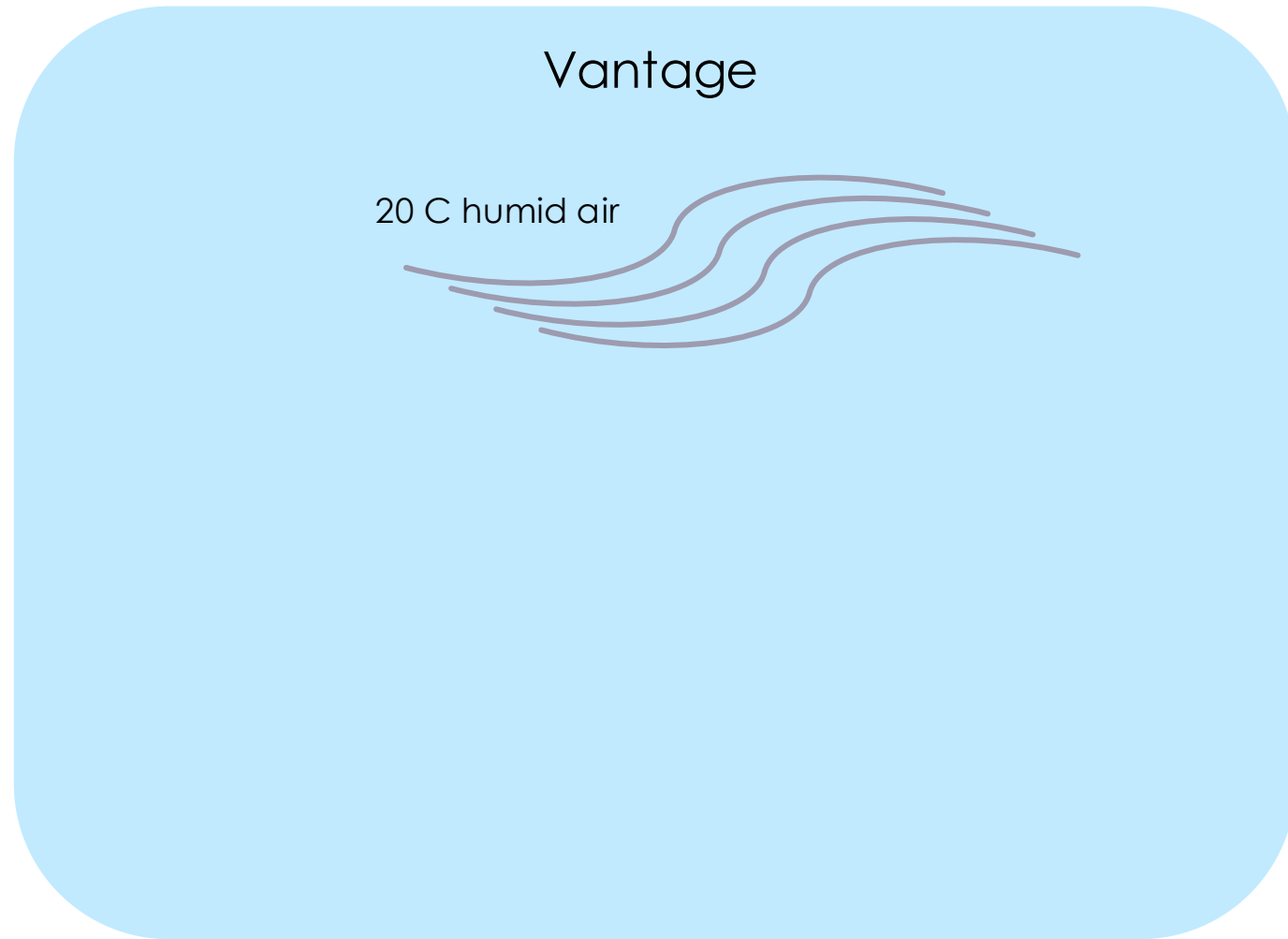
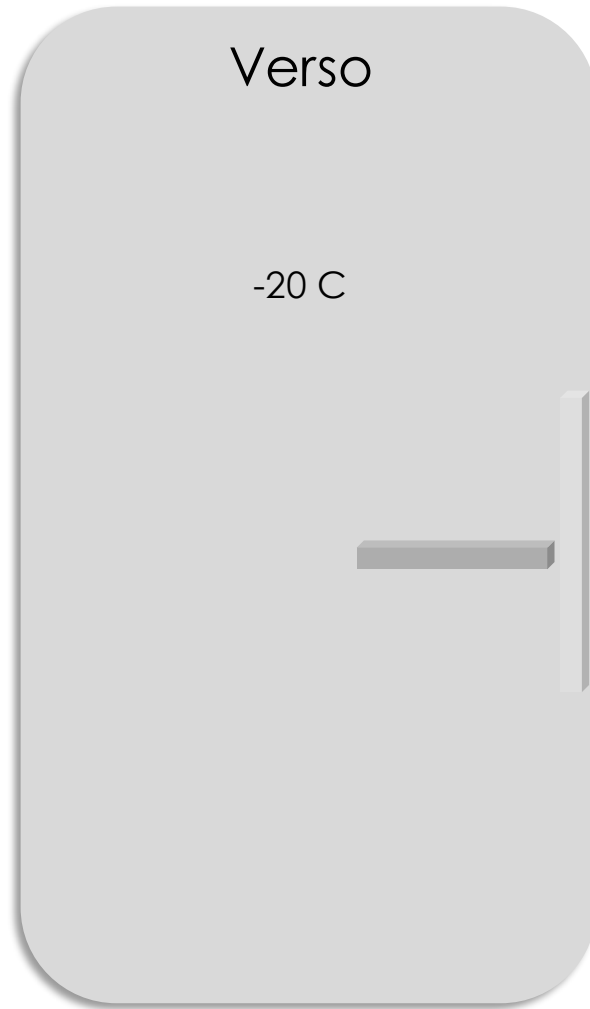
1. Hamilton Hand-off Load Bearing



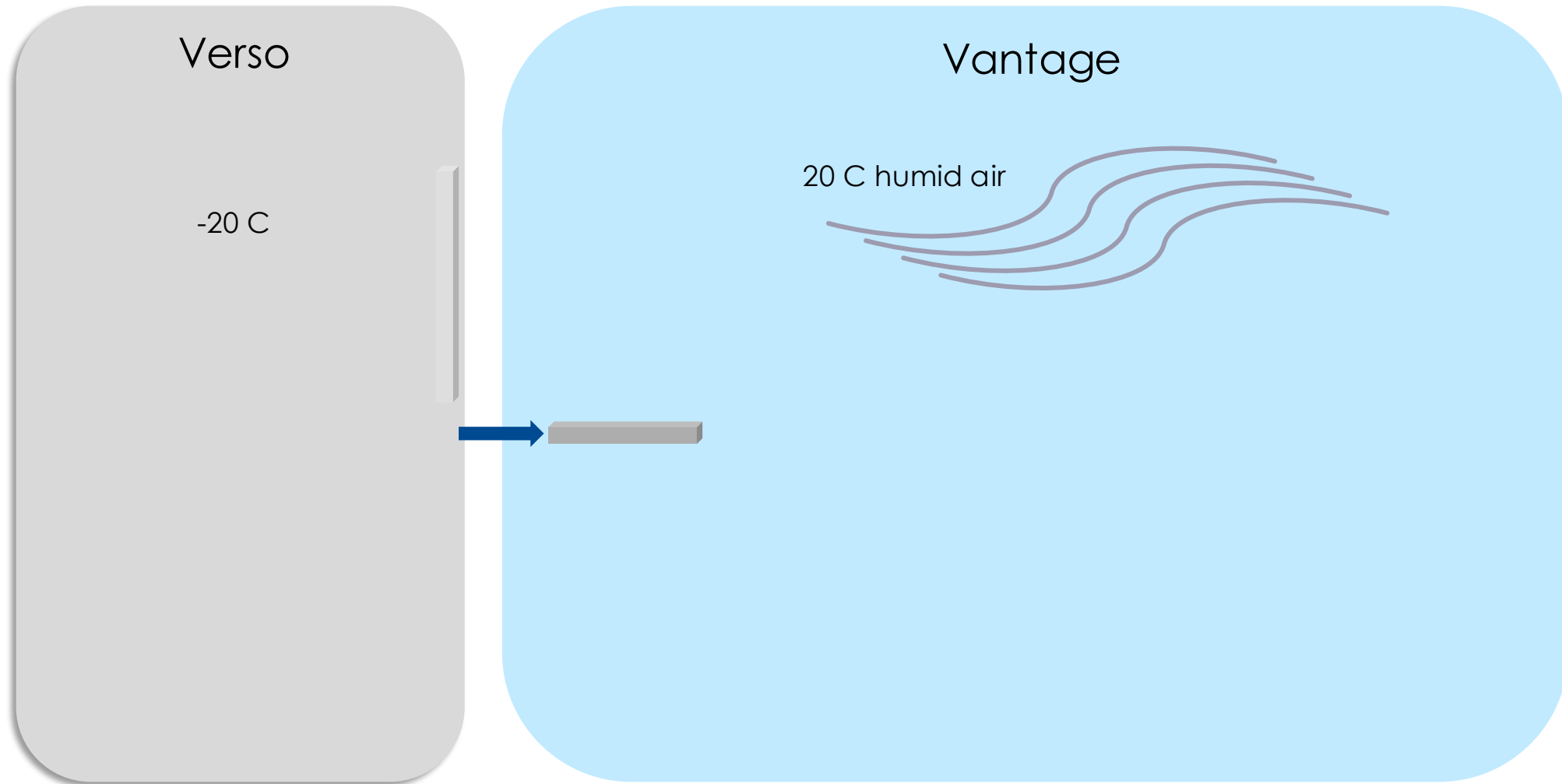
1. Hamilton Hand-off Load Bearing



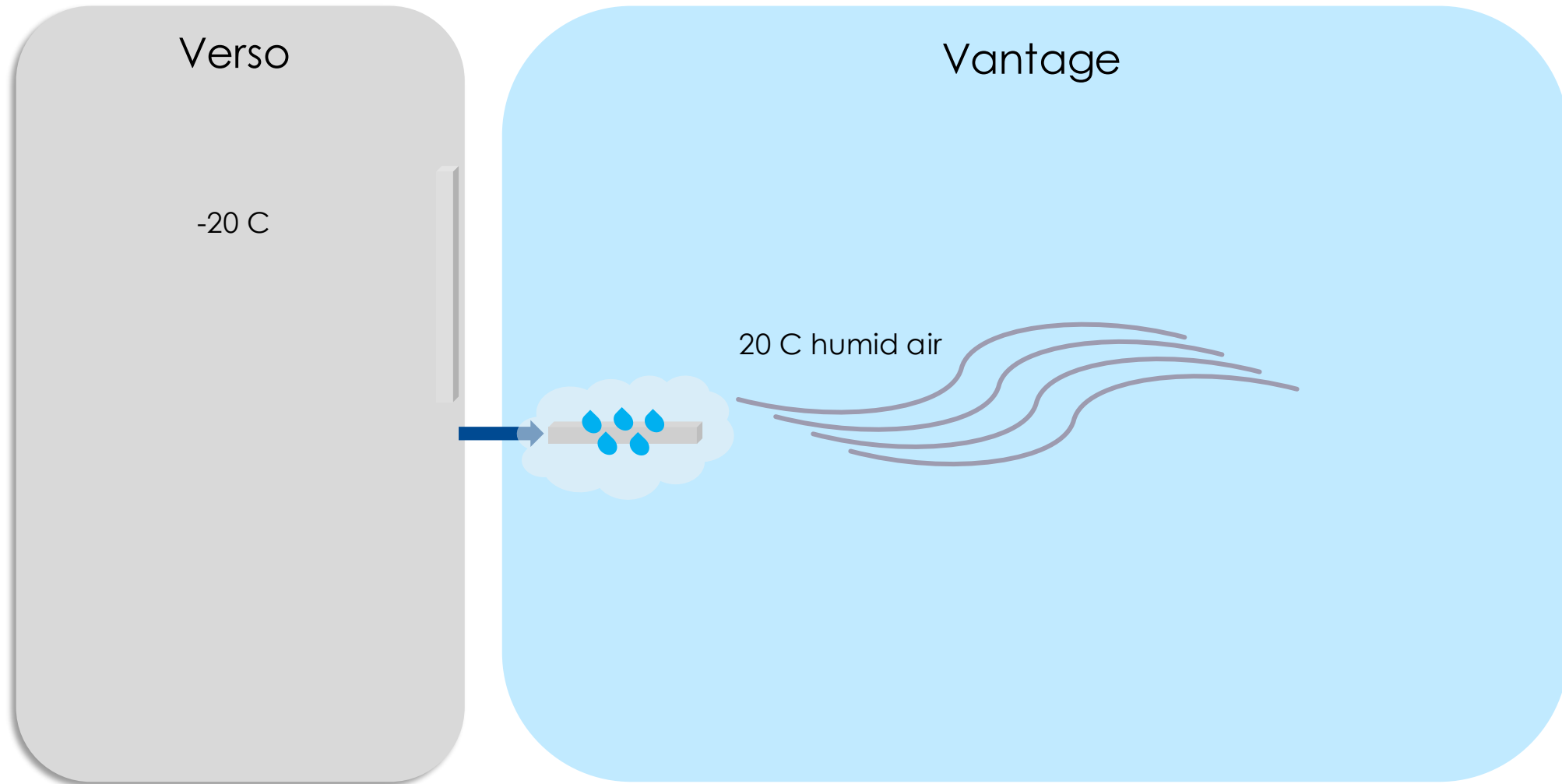
2. Moisture Reduction



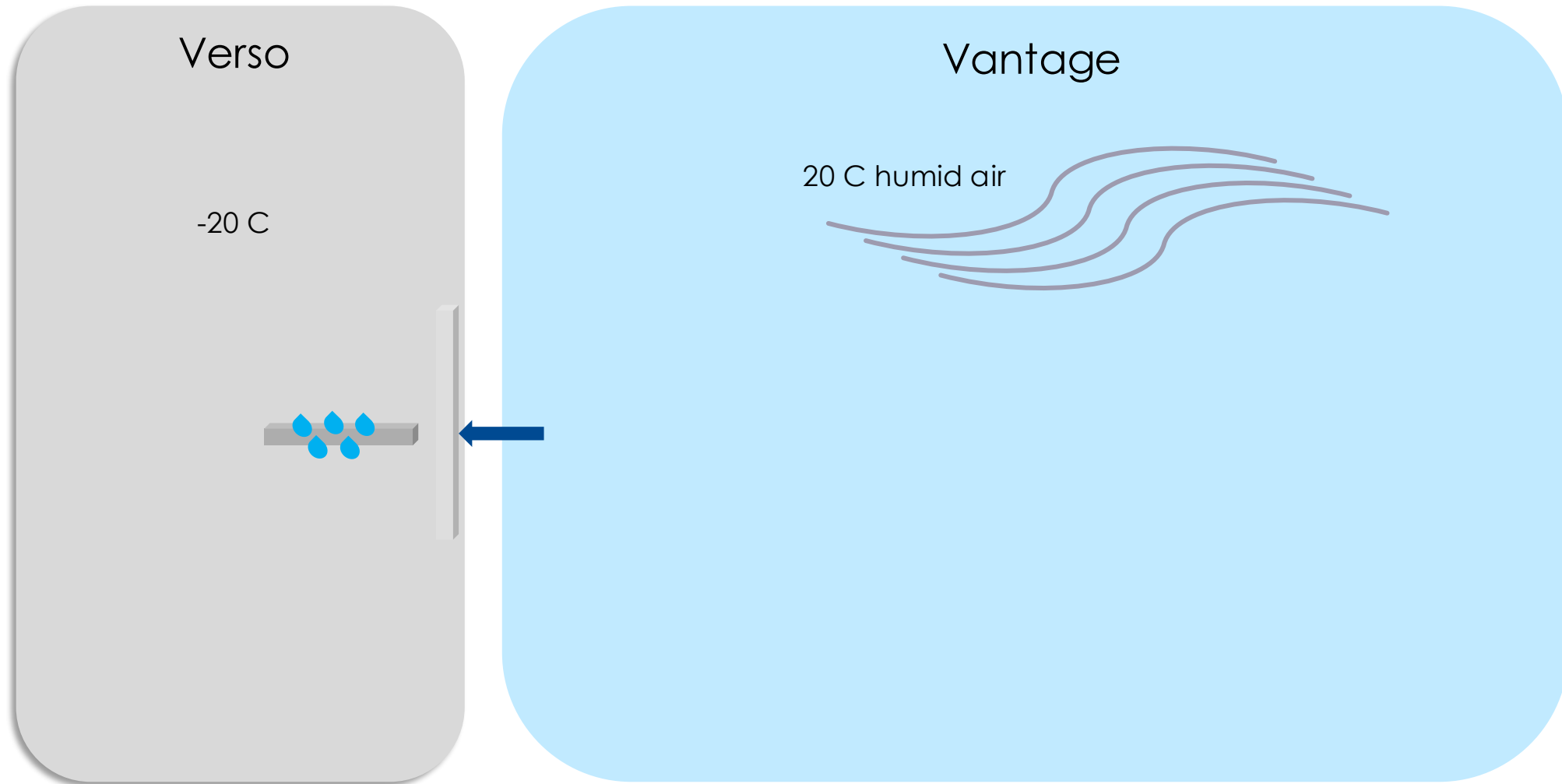
2. Moisture Reduction



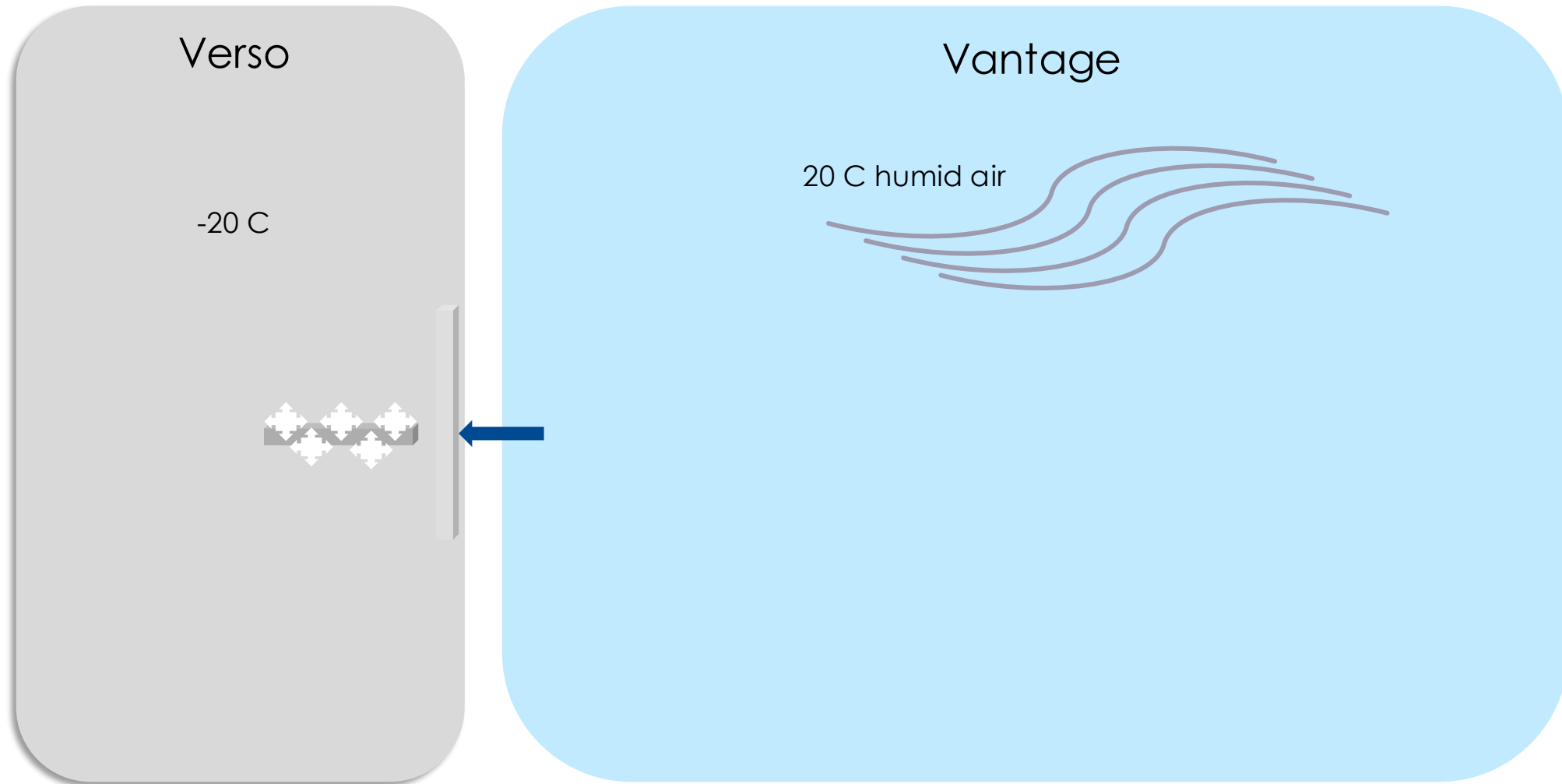
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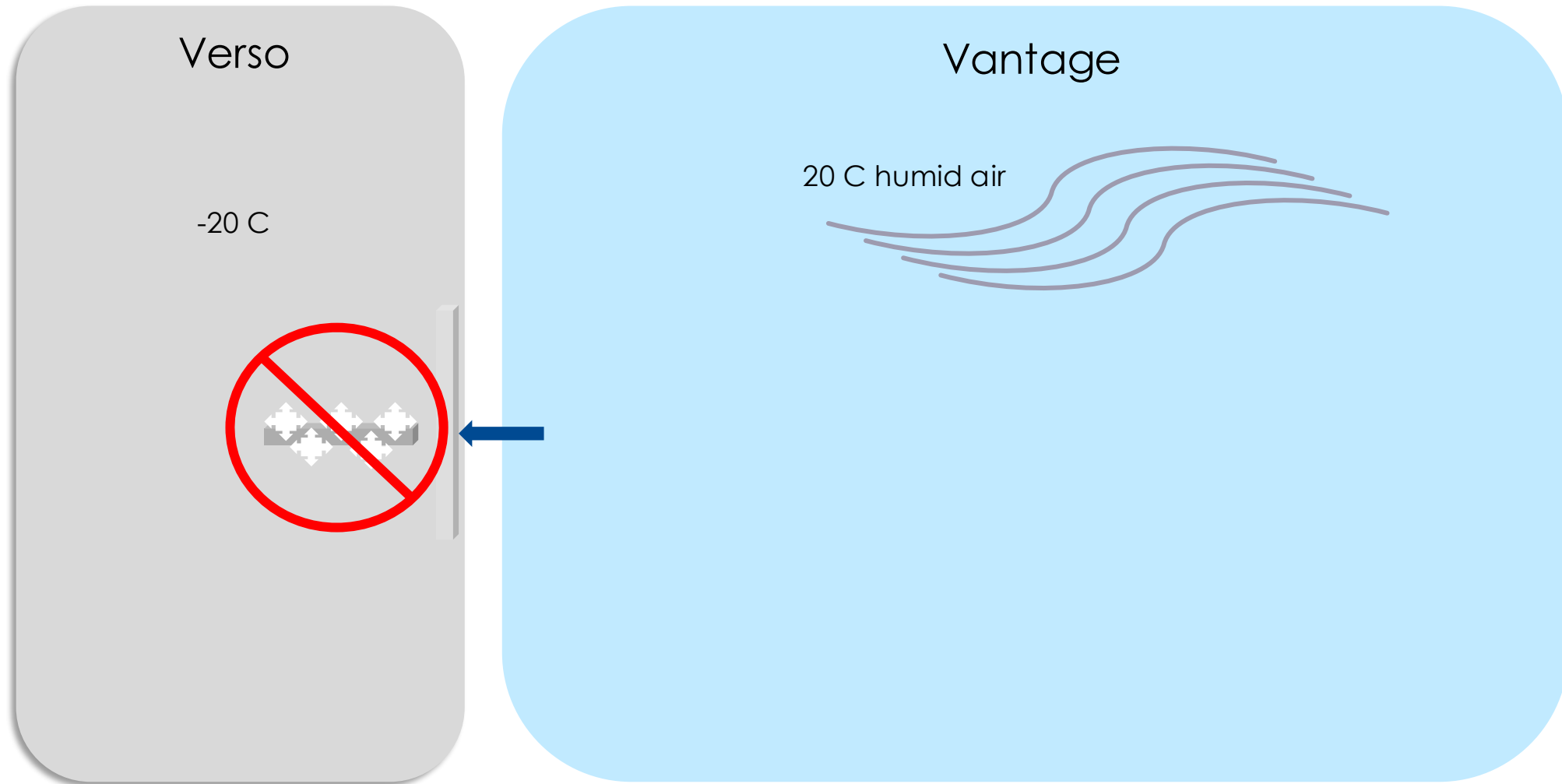
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250/02

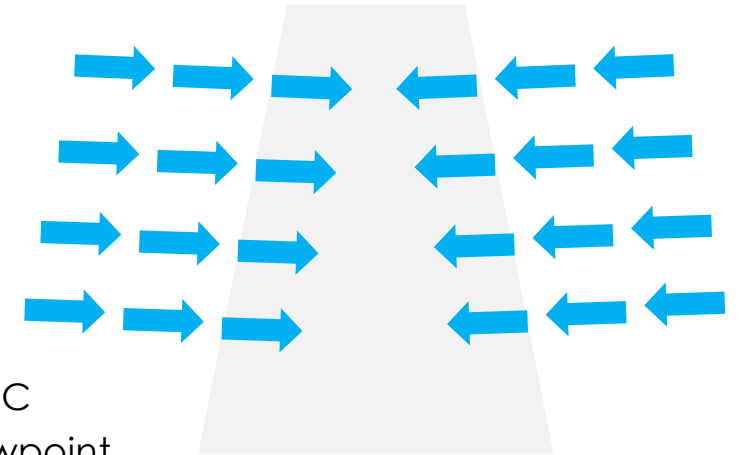
s/o

2. The Design



3D printed Polyamide

Vents above the hand blow
air down -> air curtain



-40 C
dewpoint

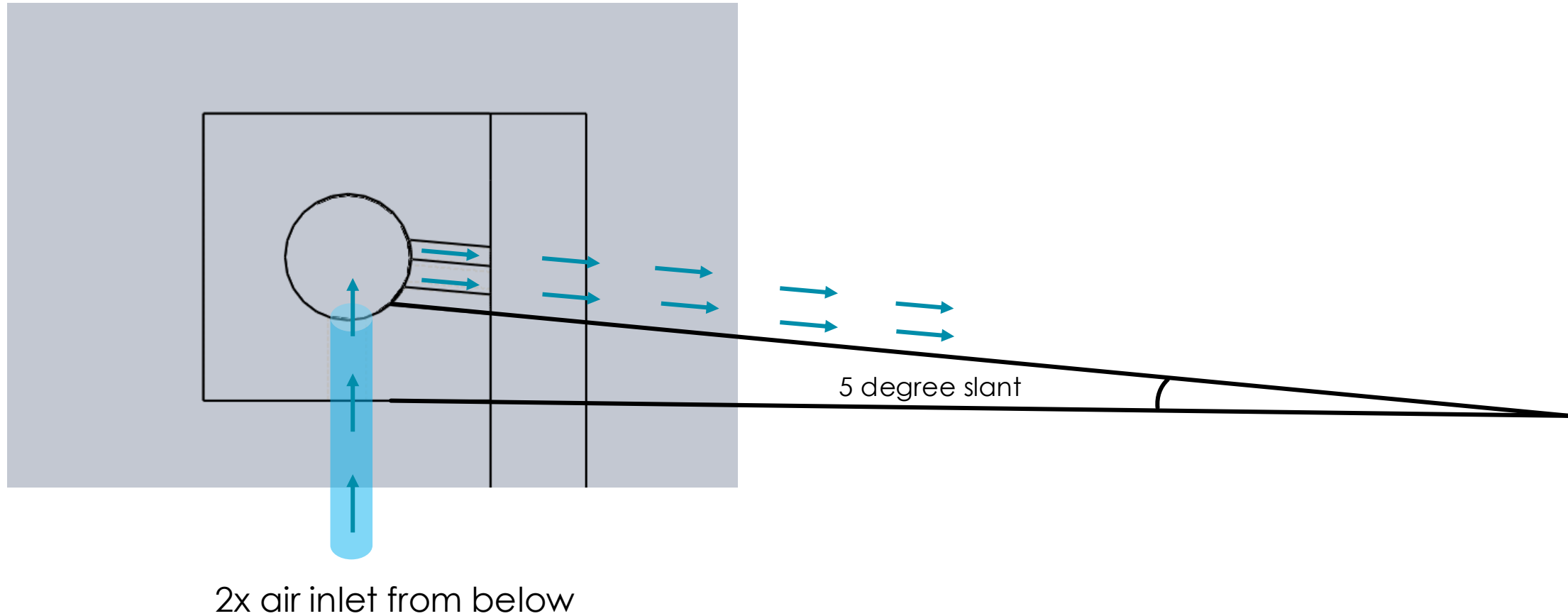
2. Air flow functional requirements

Variable	Specification Limit	Why?	Value
Reynold's Number	<2000	We want laminar flow	532
Pressure	< 6 bar	Max pressure of the compressed air flow in the lab	$3.8 \cdot 10^{-4}$ bar
Flow Rate	<830 l/min	Financial reasons	385 L/min

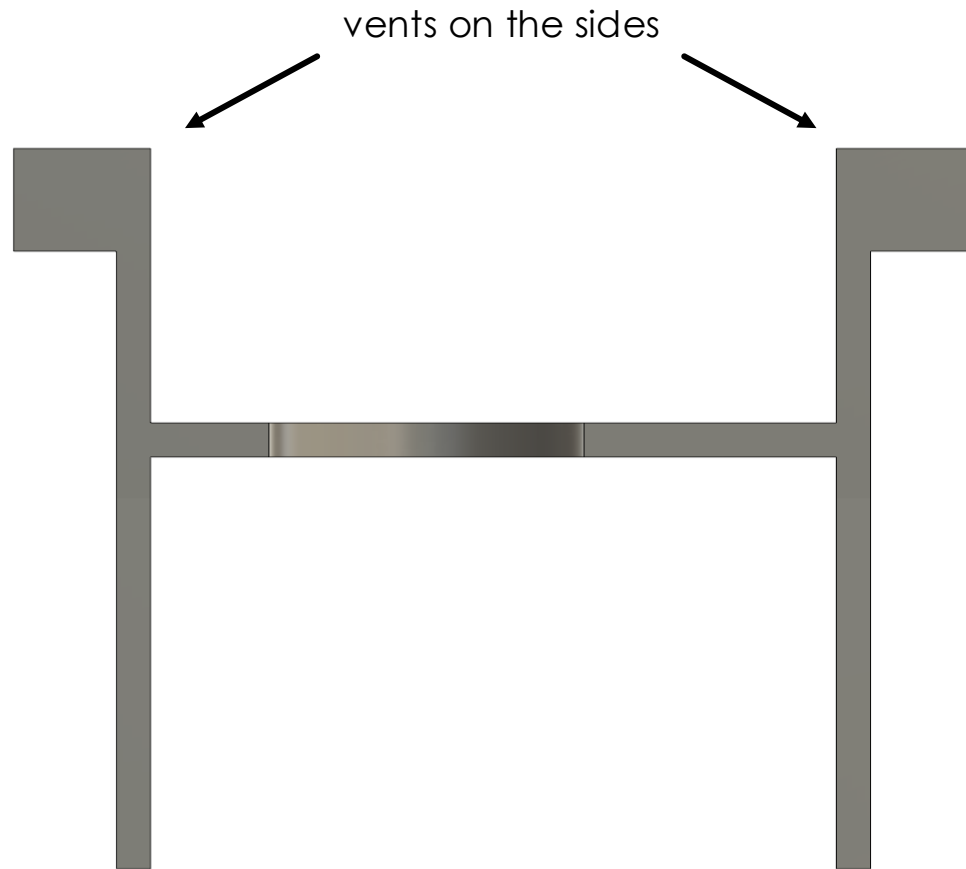
2. Air flow design – drawing the vents



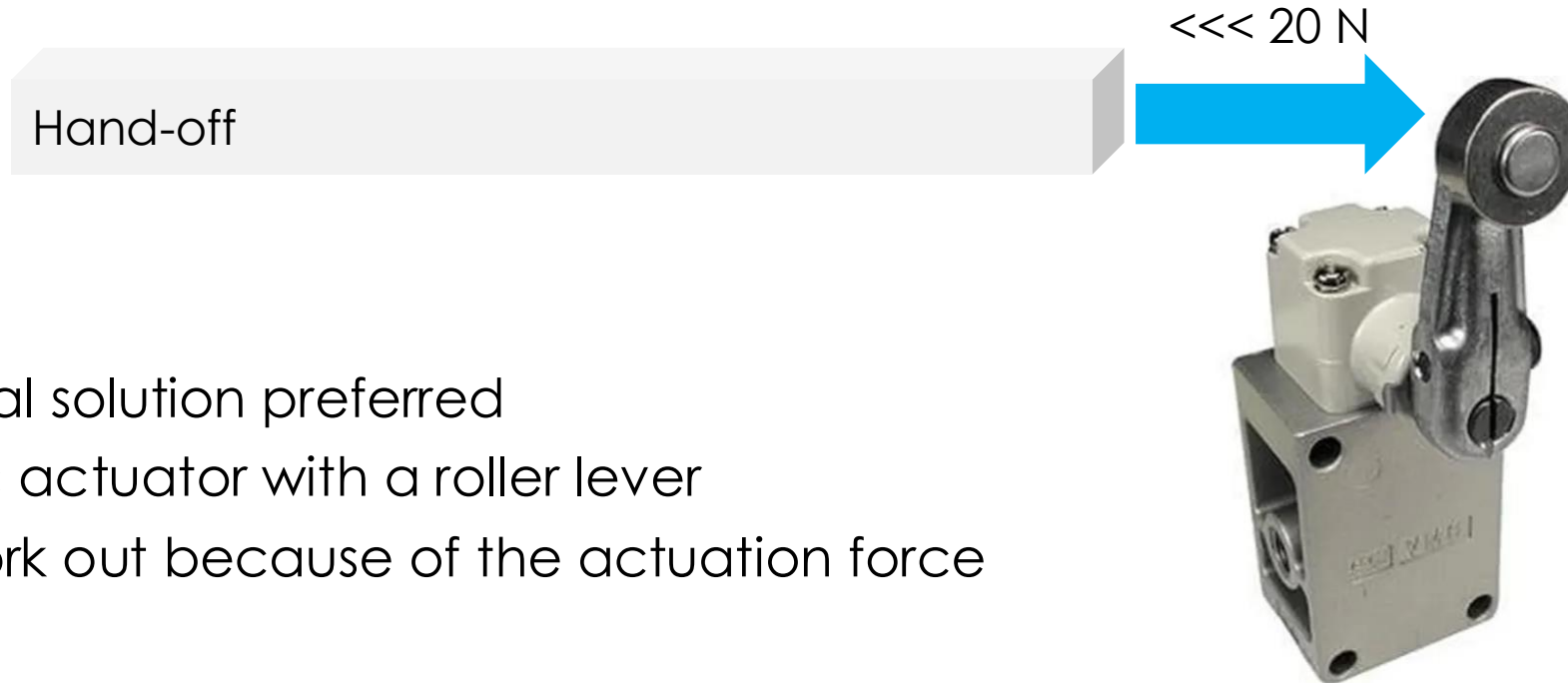
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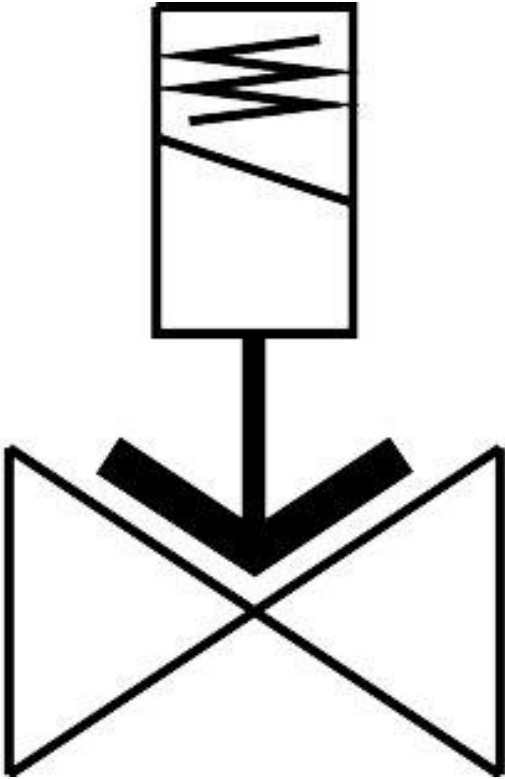


2. Air flow design – choosing the valve



- Mechanical solution preferred
- Pneumatic actuator with a roller lever
- Did not work out because of the actuation force

2. Air flow design – choosing the valve



2/2 valve:

- 2 ports: inlet and outlet for air
- 2 positions: open and closed air flow

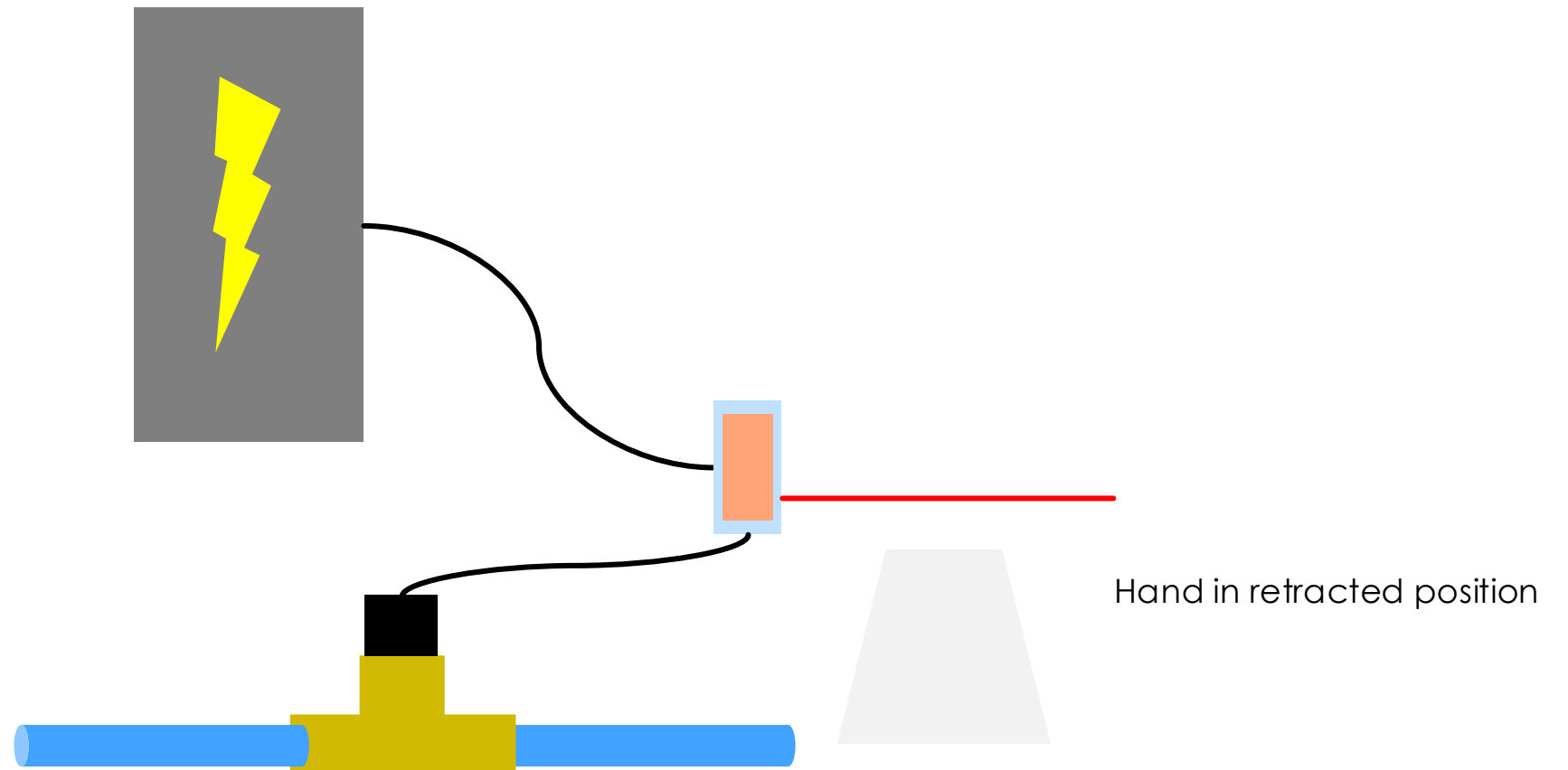
Solenoid valve: uses an electrical signal to know when to change positions

Nominal flow rate = 1920 l/min

G1/4 connections, use festo push in to 8mm tubing

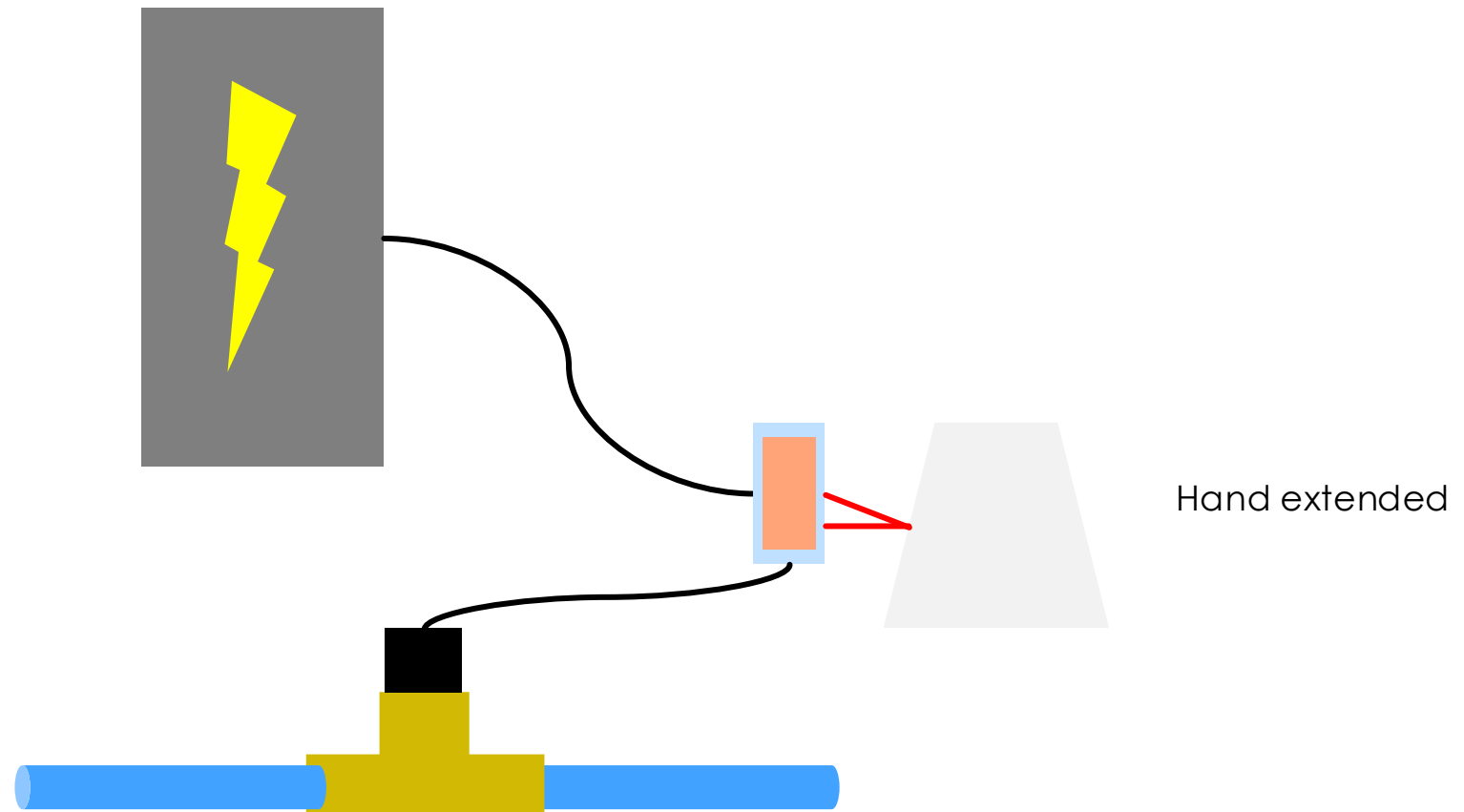
2. Automation

Wall power -> 24 VDC power supply -> photoelectric sensor -> solenoid valve ON



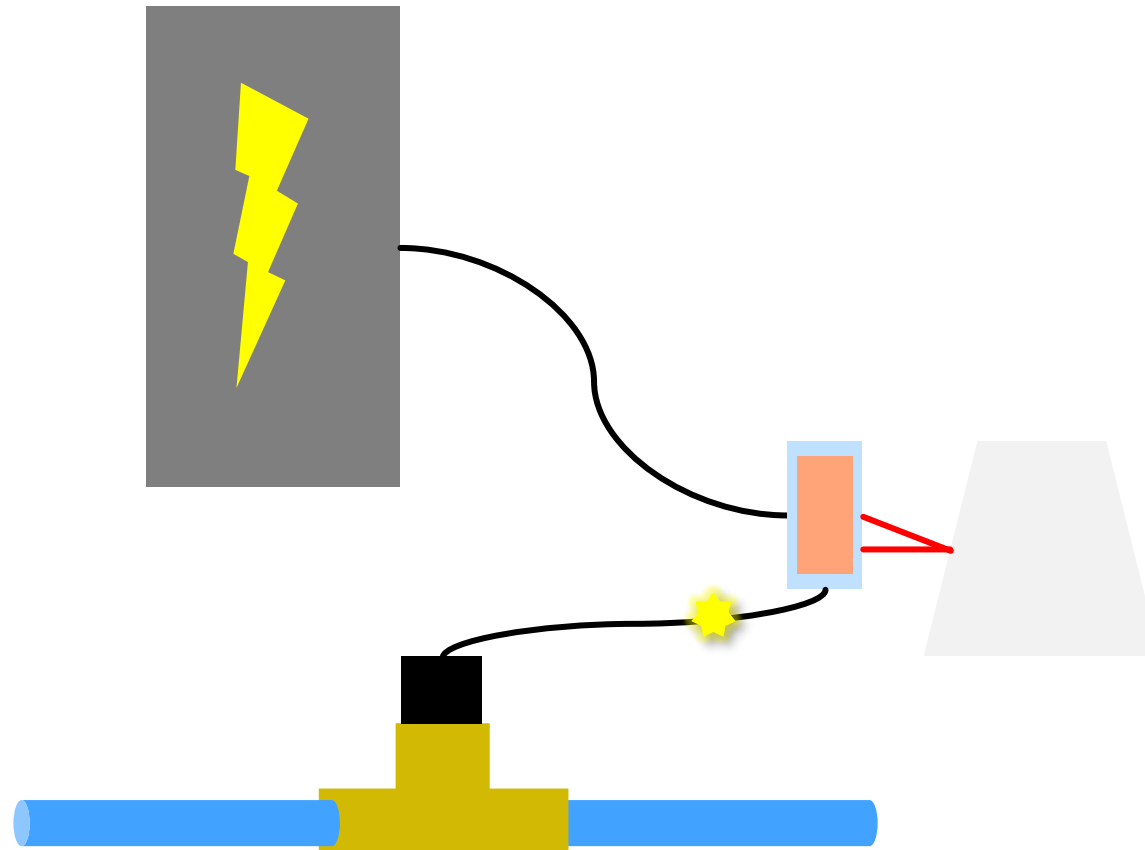
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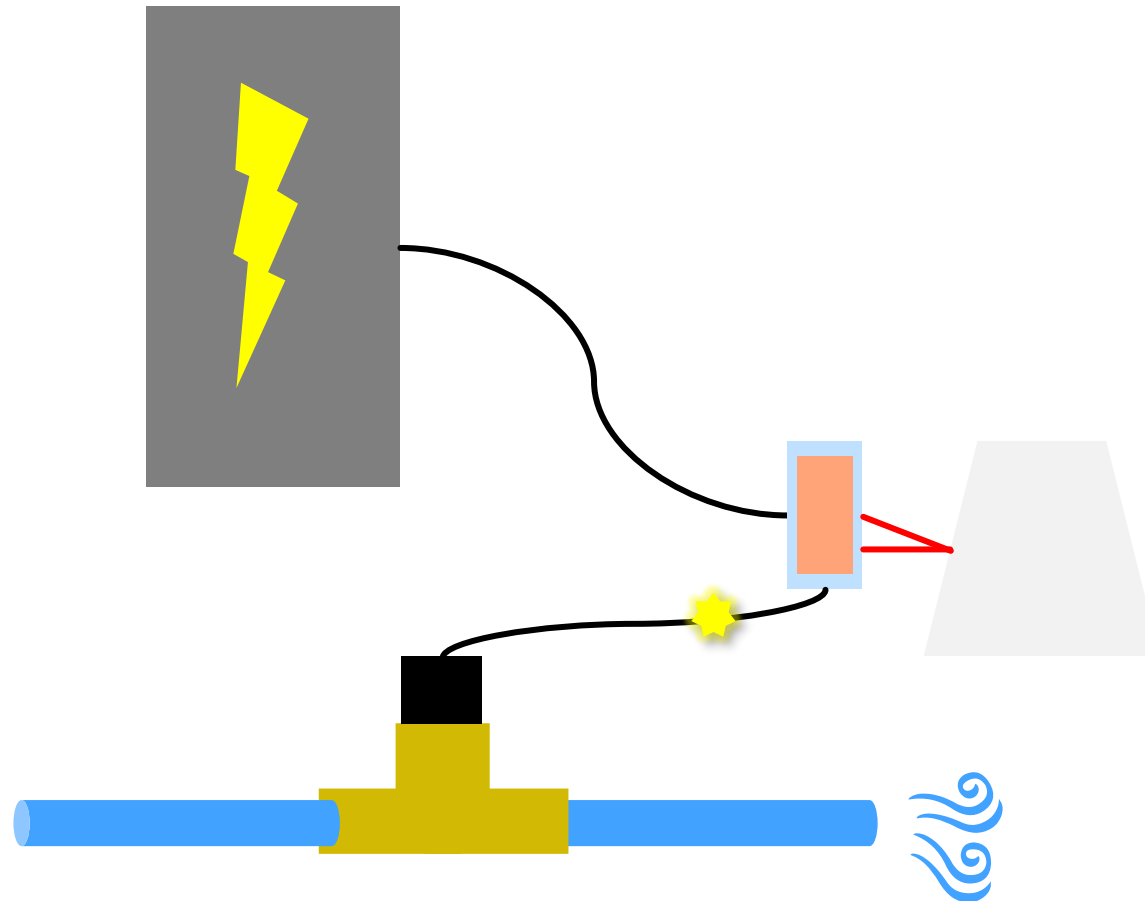
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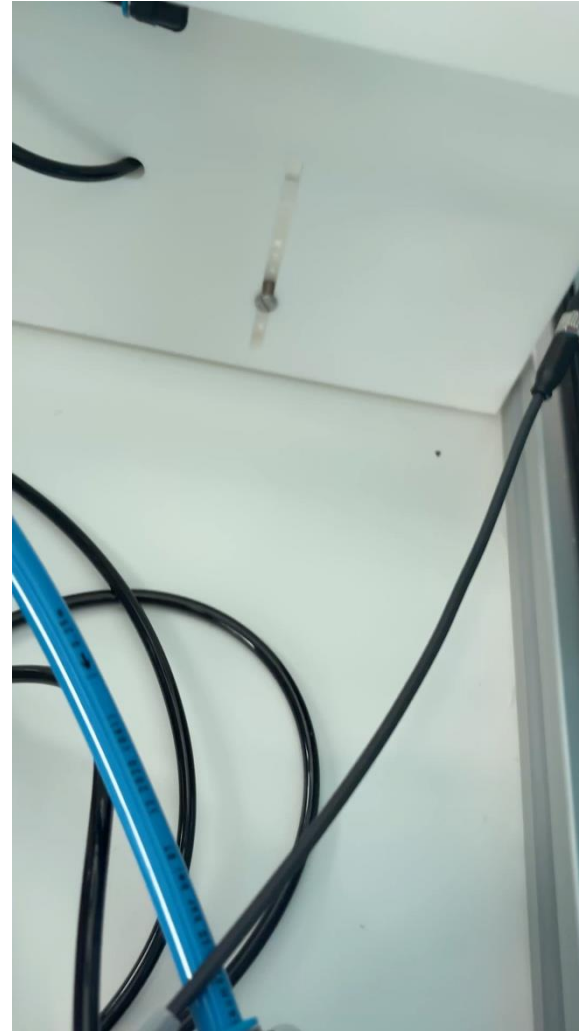
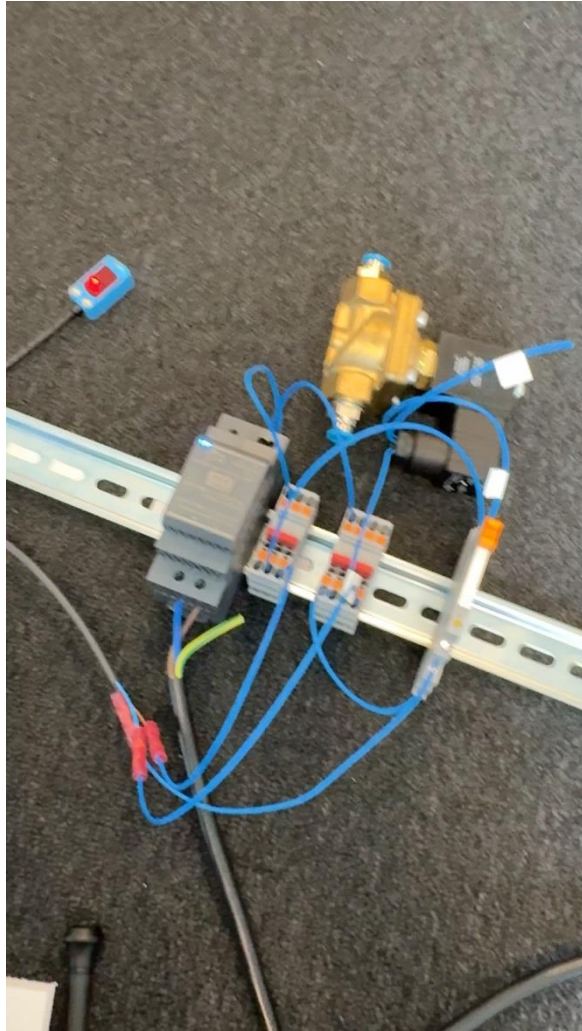


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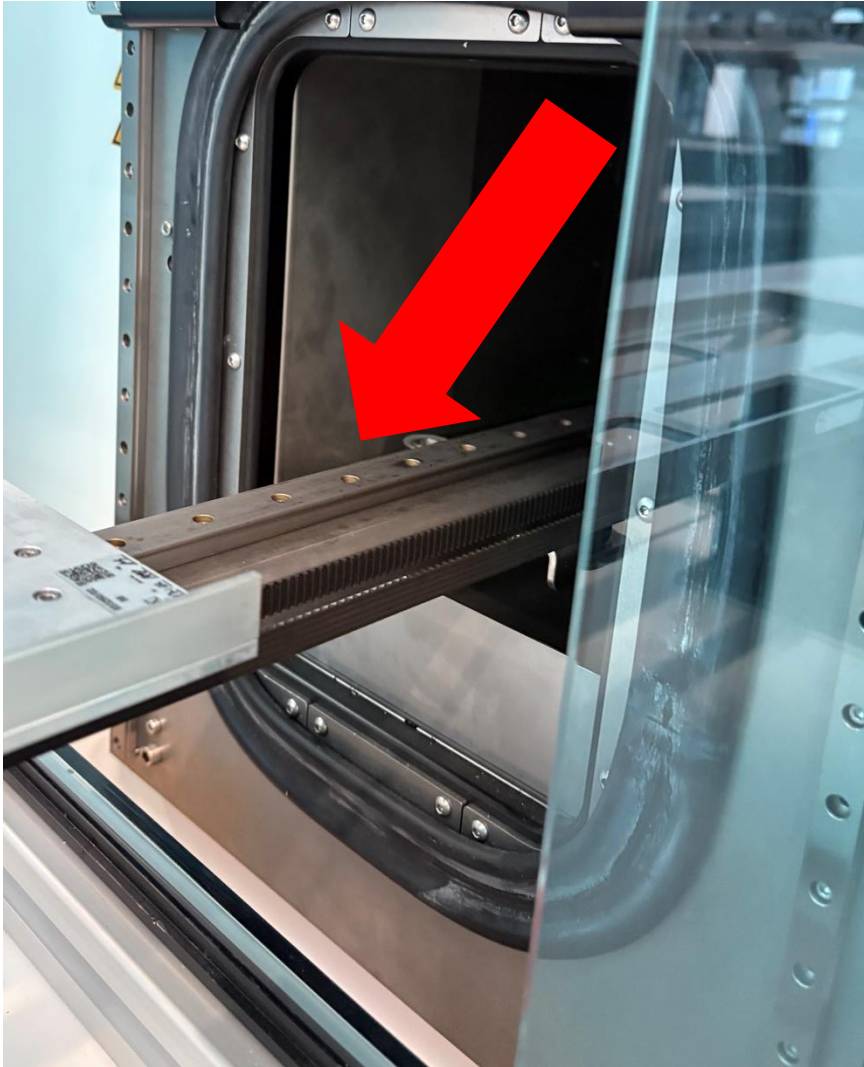
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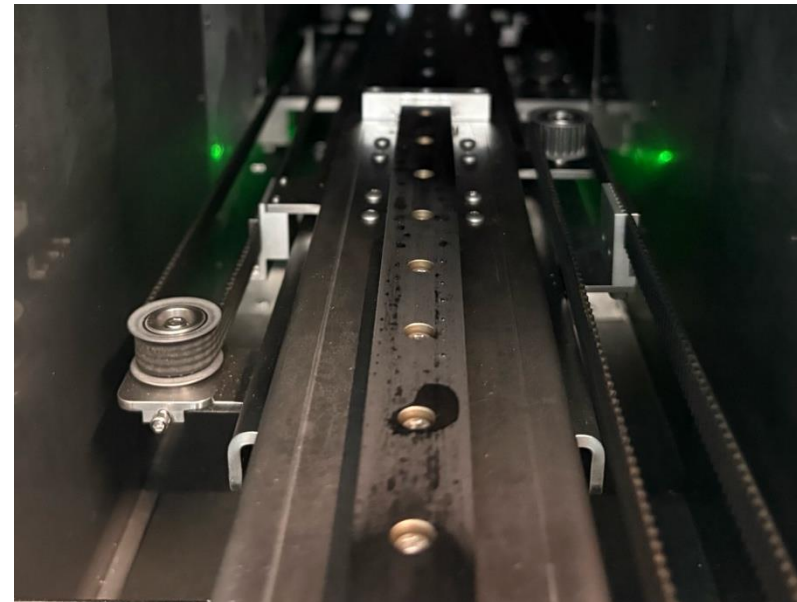
2. Automation – circuit + sensor



3. Moisture Reduction Extension Module

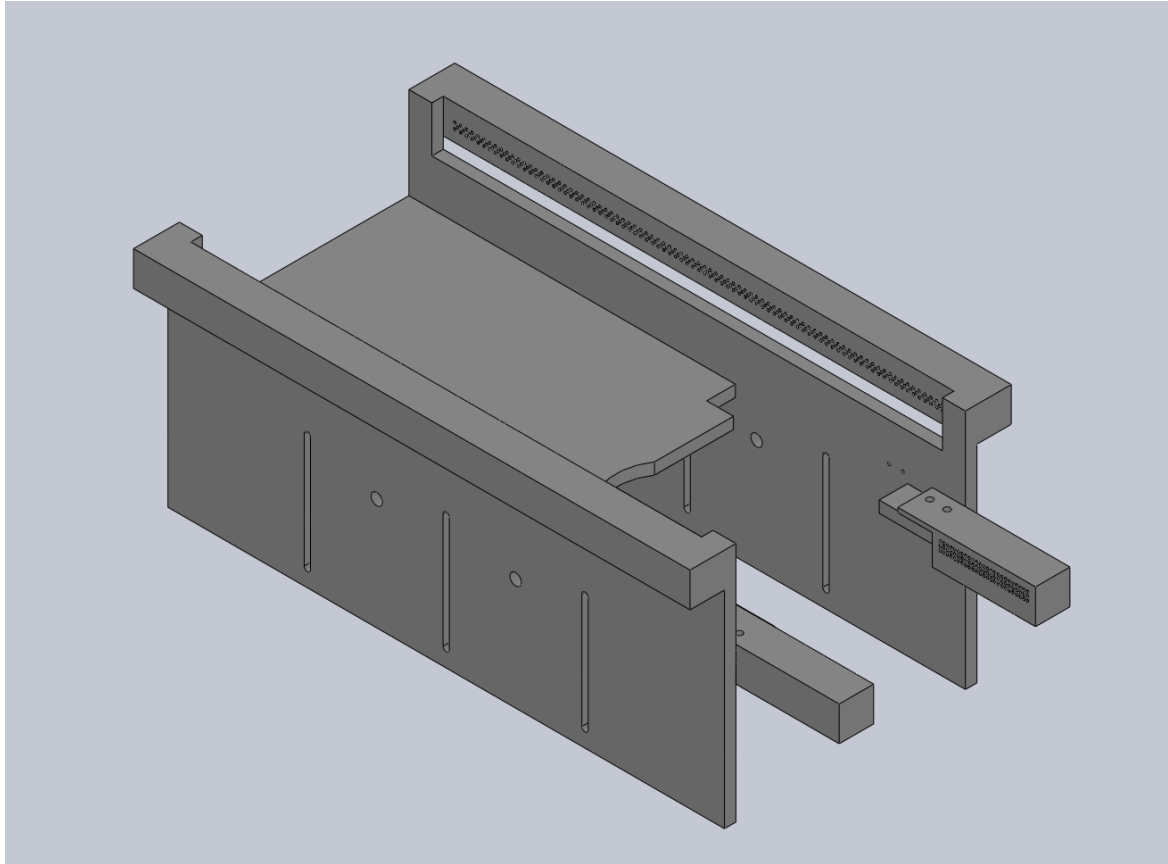


- Moisture forms here between the gap of the two instruments
- Original design did not account for condensation here, so need to extend air flow

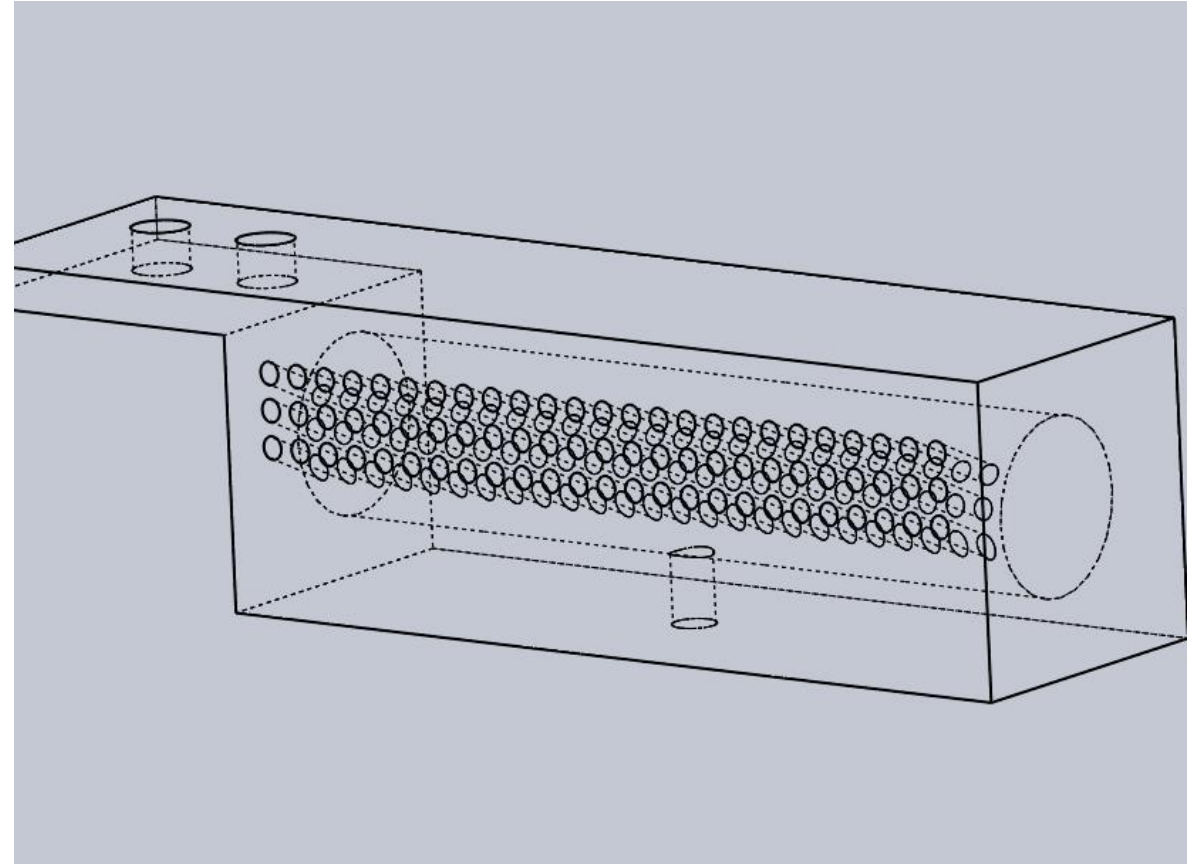


condensation on arm

3. Moisture Reduction Extension Module



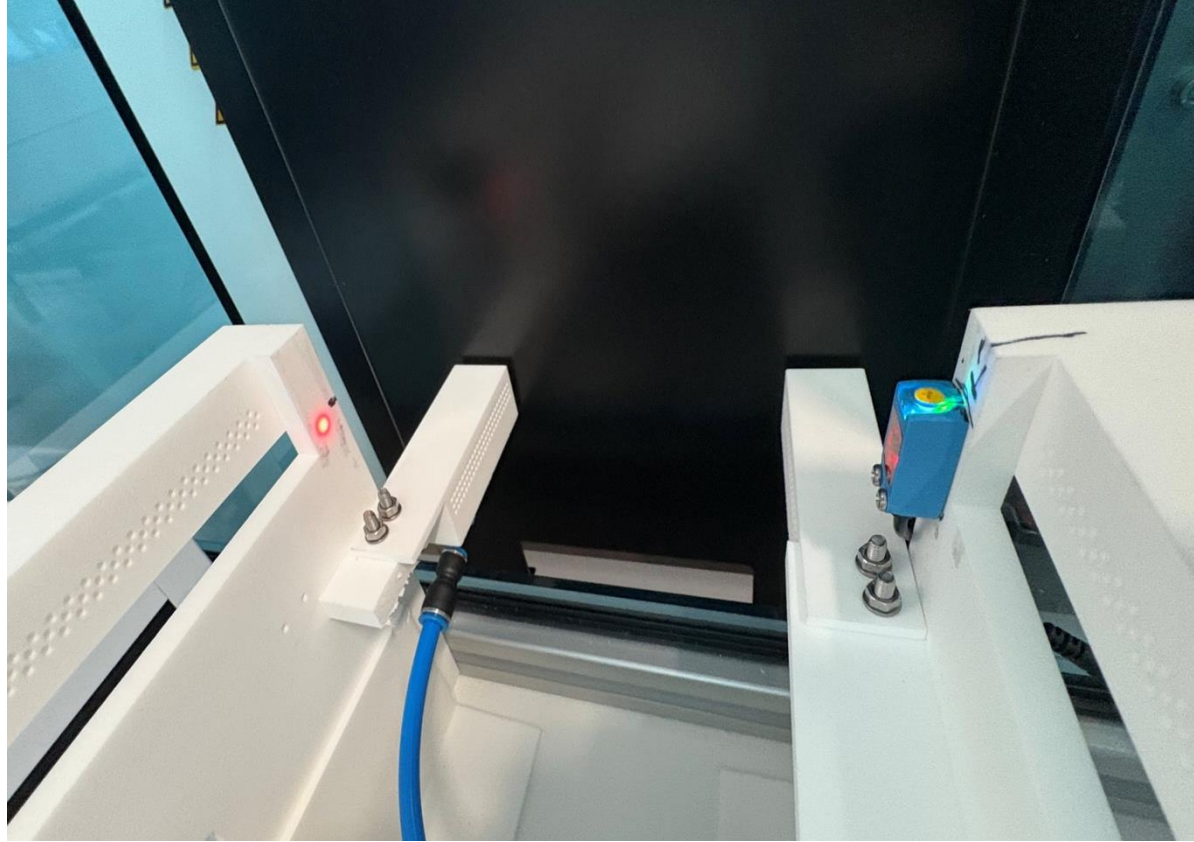
25 degree slant upwards



Real World System



Hand in front of sensor



Sensor shining laser, extension modules attached



Top-down view of system

Results: Noise Levels

Results show we don't need ear protection to work in lab while air is on

Flow Rate (L/min)	Noise Level Behind Vantage (dB) (door open)	Noise Level Front of Vantage (dB) (door open)	Noise Level Front of Vantage (dB) (door closed)
200	87-90	64-65	63-65
250	89-92	69-72	63-67
300	90-92	70-72	65-68
350	90-94	72-73	67-68
400	93-97	73-76	68-71

Danish noise level standards:
Continuous noise > 85 dB needs hearing protection.

Greater than this requires reduced exposure.

88 dB(A) for 4 hours

91 dB(A) for 2 hours

94 dB(A) for 1 hour

97 dB(A) for 30 minutes

Loud room comparison: 80 dB

Disclaimer: Phone app was used for the measurements, may not be accurate

Results: Hand is Dry Before Retracting

Time To Dry Throughout the Day vs. Flow Rate

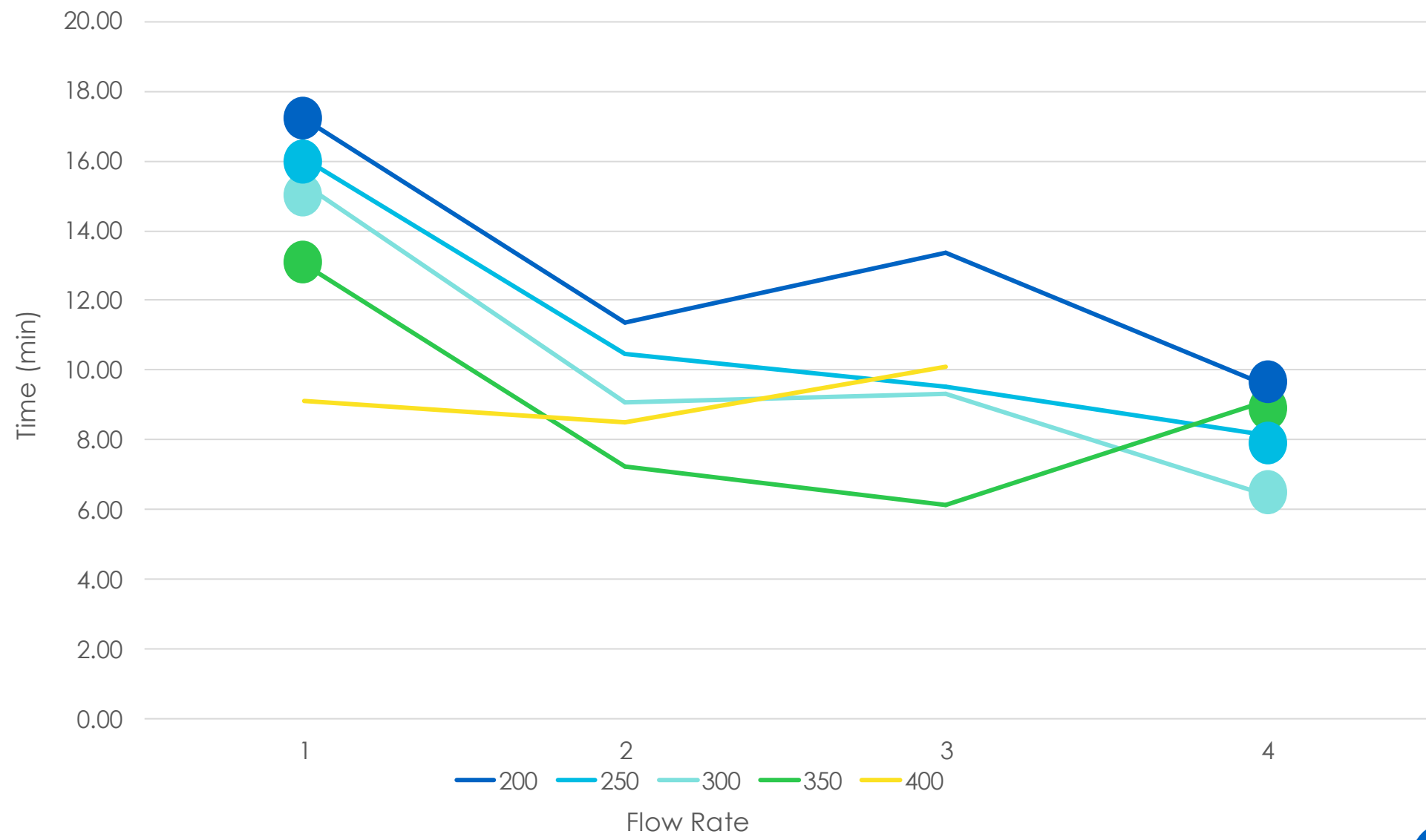
≥1 hr

≥1 hr

Flow Rate (L/min)	200	250	300	350	400
Time 1	17:21*	16:07*	15:31*	13:06*	9:11*
Time 2	11:35	10:48	9:08	7:23	8:49
Time 3	13:35	9:53	9:31	6:14	10:11
Time 4 (with extension)	9:55	8:15	6:41	9:10*	-

*first of the day, tray is coldest

Time To Dry Throughout the Day vs. Flow Rate

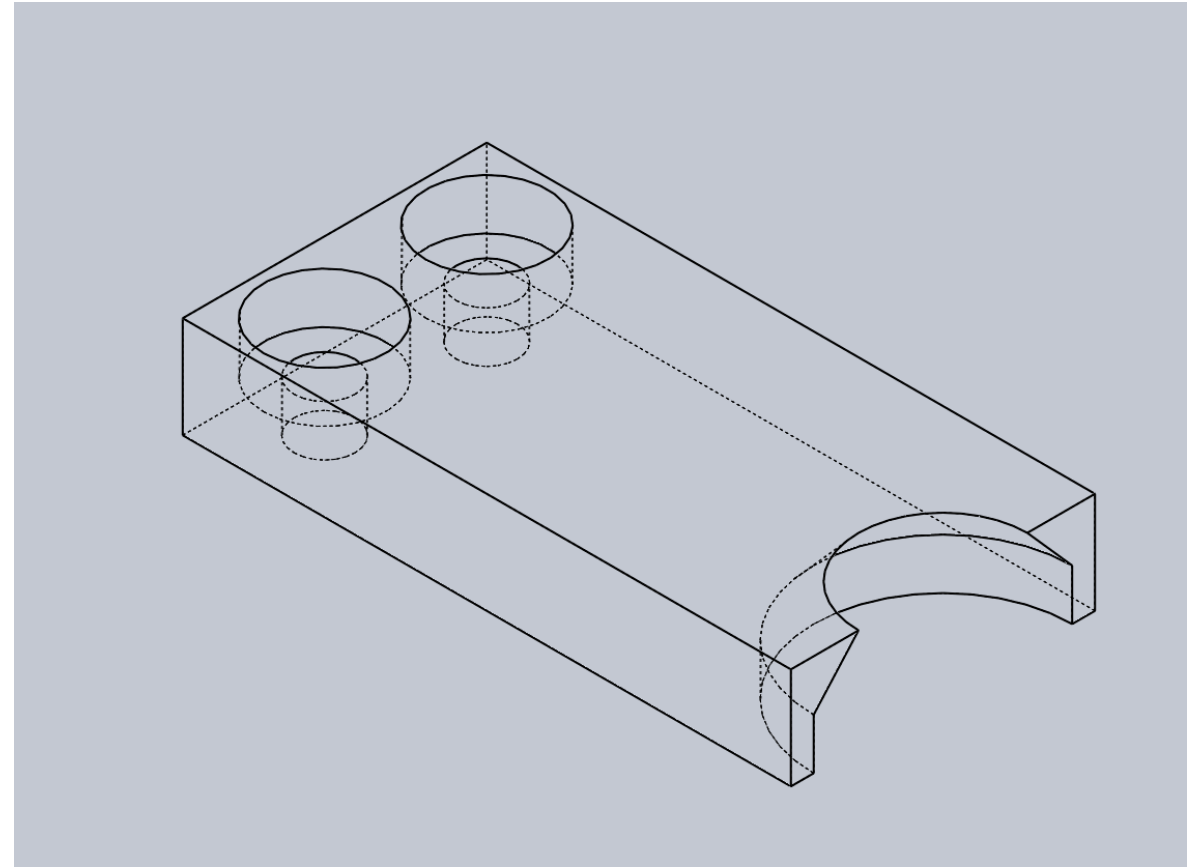
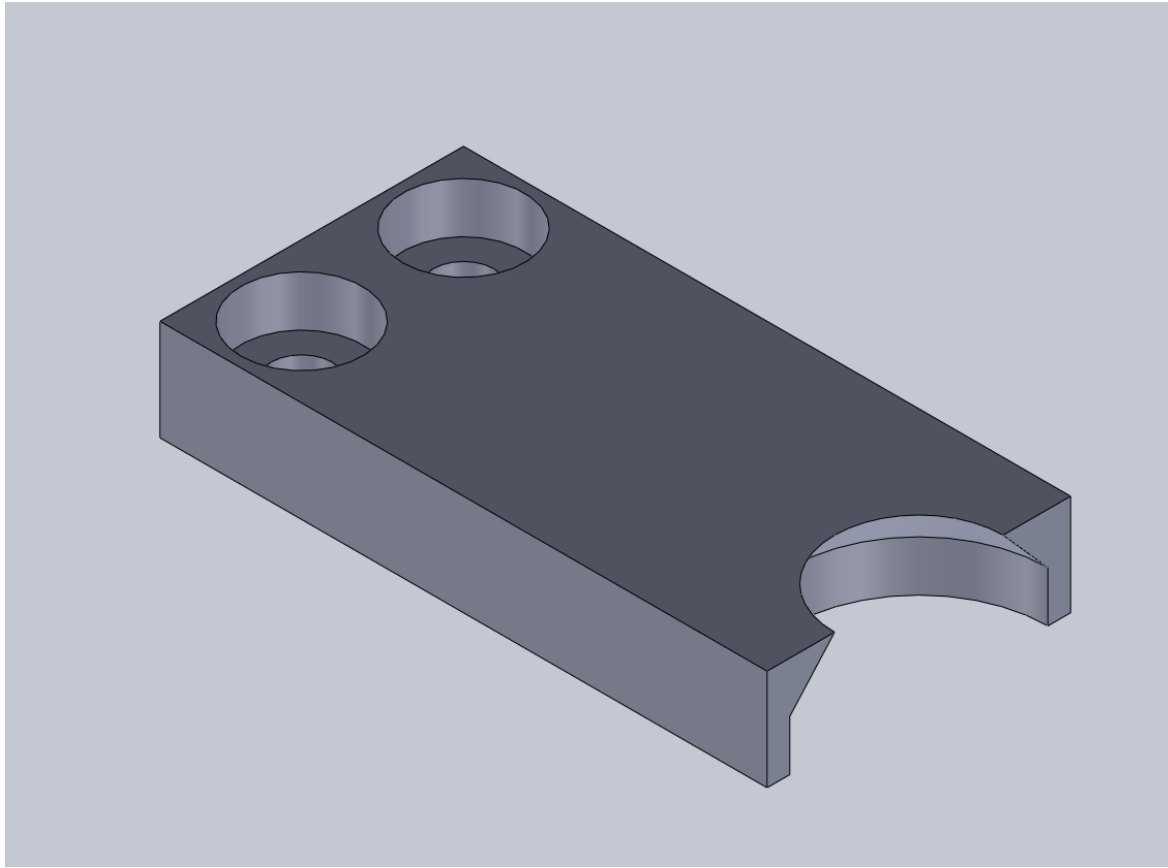


Conclusion: Overall Improvement!

- Progress made
- Design uses a different mechanism to solve the problem
- Design Changes:
 - Larger holes -> hairdryer mech
 - More inlet holes -> better mitigate pressure differences
 - Vent locations -> above vs below
- Testing Changes:
 - Full tray
 - Standardized time intervals

4. Echo Feet Mount

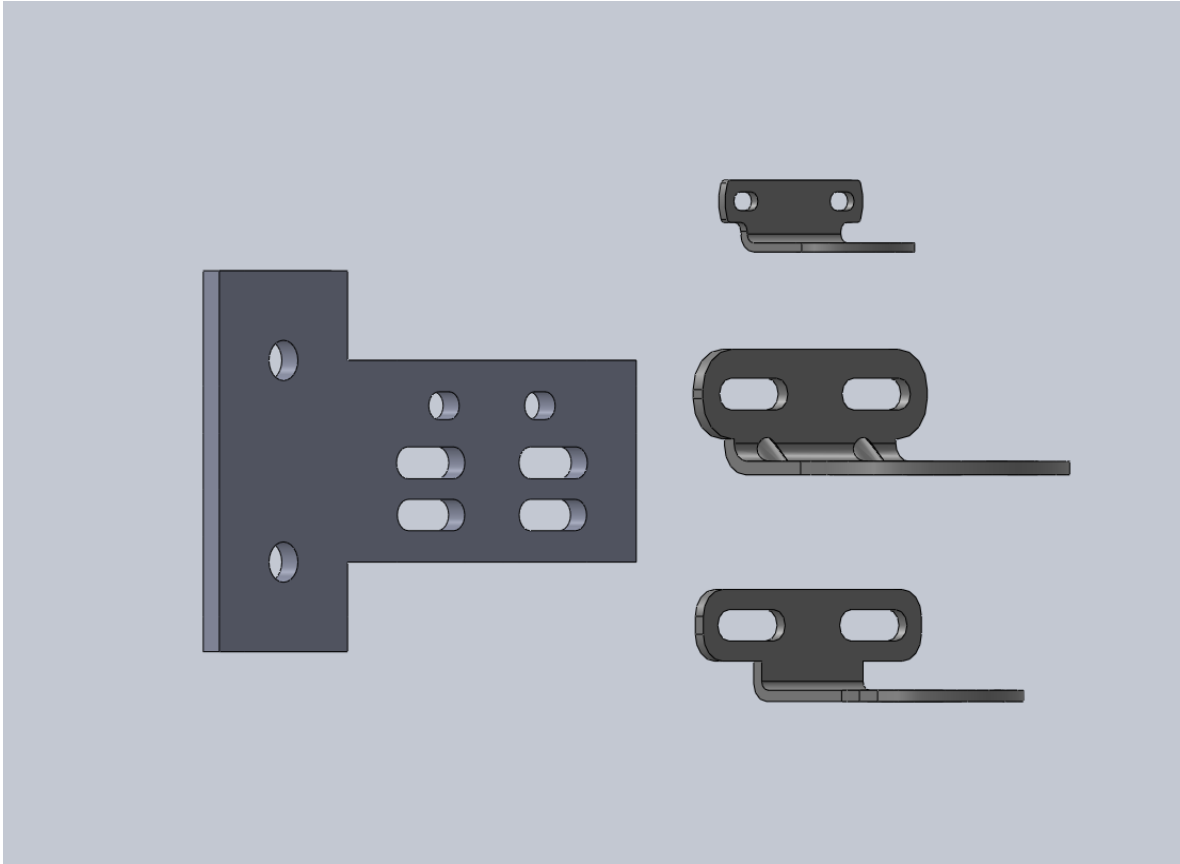
The echo at RG gets shifted around a lot, so we need to mount it without having to move it.



4. Echo Feet Mount



5. Festo Regulator Mounting



- Table legs are T-slots
- Mount to legs vertically
- Mount regulator mount horizontally
- Three different sizes of Festo mounts

Thanks for listening and a great summer!



Results: Hand is Dry Before Retracting

Flow Rate (L/min)	Pressure (bar)*	Time to frost (min)	Time to dry (min)
200	$1 < P < 1.5$	1:40	17:21
250	$1.5 < P < 2$	1:10	16:07
300	$2 < P < 2.5$	1:30	15:31
350	$2.5 < P < 3$	1:30	13:06
400	3	1:30	9:11

*pressure gauge has low resolution